

Marketing Funnel & Conversion Performance Analysis

Future Interns – Data Science & Analytics Task 3 (2026)

This project analyzes the UCI Bank Marketing dataset, containing 45,211 direct-marketing campaign records and 17 variables. The target y indicates whether a client subscribed to a term deposit.

1. Executive Summary

KPI	Result
Campaign contacts	45,211
Subscriptions	5,289
Non-conversions	39,922
Overall conversion rate	11.70%
Overall drop-off	88.30%

2. Funnel Definition

The dataset is telemarketing data, not web analytics. It has no explicit Visitor or Lead fields. Therefore the project uses the observable funnel Campaign Contacts → Subscriptions and does not invent a lead count.

3. Key Findings

- Only 11.7% of campaign contacts converted, leaving an 88.3% non-conversion share.
- Previous campaign success is a strong prioritization signal: 978 of 1,511 clients with a recorded previous success converted again (~64.7%).
- Cellular outreach converts at about 14.9% versus about 13.4% for telephone in the reconstructed full contact table.
- Students, retired and unemployed groups show relatively high observed subscription rates.
- March and December show high observed rates, but their contact volumes are much smaller than May–August.
- Published analysis reports a general downward conversion trend as current-campaign contact count increases, consistent with possible contact fatigue.

4. Recommendations

- Prioritize previous-success customers — 978 of 1,511 clients with a recorded previous success converted again (~64.7%).
- Prefer stronger-performing contact channels — Cellular is ~14.9% versus ~13.4% for telephone in the full reconstructed contact table.
- Control repeat-contact frequency — Published analysis shows conversion generally declines as current-campaign contact count rises, suggesting contact fatigue.
- Use seasonal timing carefully — March and December show high observed rates, but volumes are much smaller than May–August.
- Use segment-based targeting — Students, retired and unemployed groups show relatively high observed subscription rates.
- Validate with experiments — Associations are not causal proof; A/B tests should confirm campaign changes.

5. Data Dictionary

Column	Meaning
age	Customer age
job	Type of job
marital	Marital status
education	Education level
default	Whether customer has credit in default
balance	Average yearly balance in euros
housing	Whether customer has a housing loan
loan	Whether customer has a personal loan
contact	Communication type
day	Last contact day of month
month	Last contact month
duration	Last contact duration in seconds
campaign	Number of contacts during current campaign
pdays	Days since previous campaign contact; -1 means not previously contacted
previous	Number of contacts before current campaign
poutcome	Previous campaign outcome
y	Whether client subscribed to a term deposit

6. Analytical Cautions

- 'Unknown' is a category in the source data and is not automatically equivalent to missing.
- UCI warns that duration is known only after the call; it should not be used for realistic pre-call targeting.
- The dataset is historical and specific to a Portuguese bank campaign; findings should not automatically be generalized.
- Segment differences are associations, not causal effects. Use controlled experiments before changing campaign policy.

Source: UCI Machine Learning Repository – Bank Marketing, DOI 10.24432/C5K306.